Exploratory Data Analysis of the Instagram Reach Dataset

EDA, feature engineering, and interpretable classification for high-engagement Instagram posts

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1 Project Objective

This project analyzes a dataset of 100 Instagram posts to identify the variables most associated with high engagement. The analysis explores the role of follower count and posting age in determining whether a post receives an above-average number of likes, and evaluates this relationship through a logistic regression classifier.

Variables explored: `Followers` , `Hours.since.posted` , `Likes` , along with engineered categorical features `time` , `like` , and `high_like` .

Modeling task: Binary classification — predicting whether a post falls in the high-engagement category based on follower count and time since posting.

2 Why This Project Matters

Engagement prediction is a practical problem for anyone managing social media content. Understanding which factors drive higher engagement allows content creators and analysts to make data-informed decisions about posting strategy. This project demonstrates that simple, interpretable models built on a minimal feature set can already reveal meaningful patterns — a useful baseline before investing in more complex approaches. It also illustrates why thorough exploratory analysis before modeling is important: distributional asymmetries, class imbalance, and variable relationships all affect how results should be interpreted.

3 Executive Summary

Dataset: 100 Instagram posts, 6 variables after cleaning. **Goal:** Identify engagement patterns and predict high-engagement posts. **Key findings:** Posting age (`Hours.since.posted`) is a stronger predictor than follower count. Posts in the highest time category show notably higher engagement rates. The `Followers` distribution is strongly right-skewed. **Model result:** Trained on 70 posts and evaluated on a held-out test set of 30 — accuracy 83.3%, AUC 0.727. The model outperforms a majority-class baseline (76.7% accuracy), though recall for the high-engagement class remains moderate due to class imbalance.

4 Data Inspection

The dataset contains 100 observations and 7 variables. The index column (`S.No`) was removed as it carries no analytical information, leaving 6 variables for analysis.

Variable Overview — Instagram Reach Dataset

Variable	Type	Description
USERNAME	Categorical	Instagram account identifier
Followers	Numeric	Number of followers at time of post
Caption	Text	Post caption text
Hashtags	Text	Hashtags used in the post
Hours.since.posted	Numeric	Hours elapsed since post publication
Likes	Numeric	Number of likes received

5 Feature Engineering

Two new categorical variables were derived from the numeric data:

`time` — categorizes `Hours.since.posted` into three groups (2, 3, or 4+ hours), providing a discrete representation of posting age suitable for group comparisons.

`like` — divides `Likes` into three engagement tiers using the first and third quartiles (Q1 = 19, Q3 = 46): low (below Q1), medium (Q1–Q3), and high (above Q3).

`high_like` — a binary target variable: 1 if a post falls in the top like-tier, 0 otherwise.

Distribution of Time Category Variable

Time Category	Count	Relative Frequency
2	57	0.57
3	19	0.19
4	24	0.24

Distribution of Binary Engagement Target (high_like)

Engagement Class	Count	Relative Frequency
Low/Medium Engagement	77	0.77
High Engagement	23	0.23

The target variable is imbalanced: approximately 77% of posts fall in the Low/Medium Engagement class. This imbalance is an important consideration when interpreting classification performance.

6 Descriptive Statistics

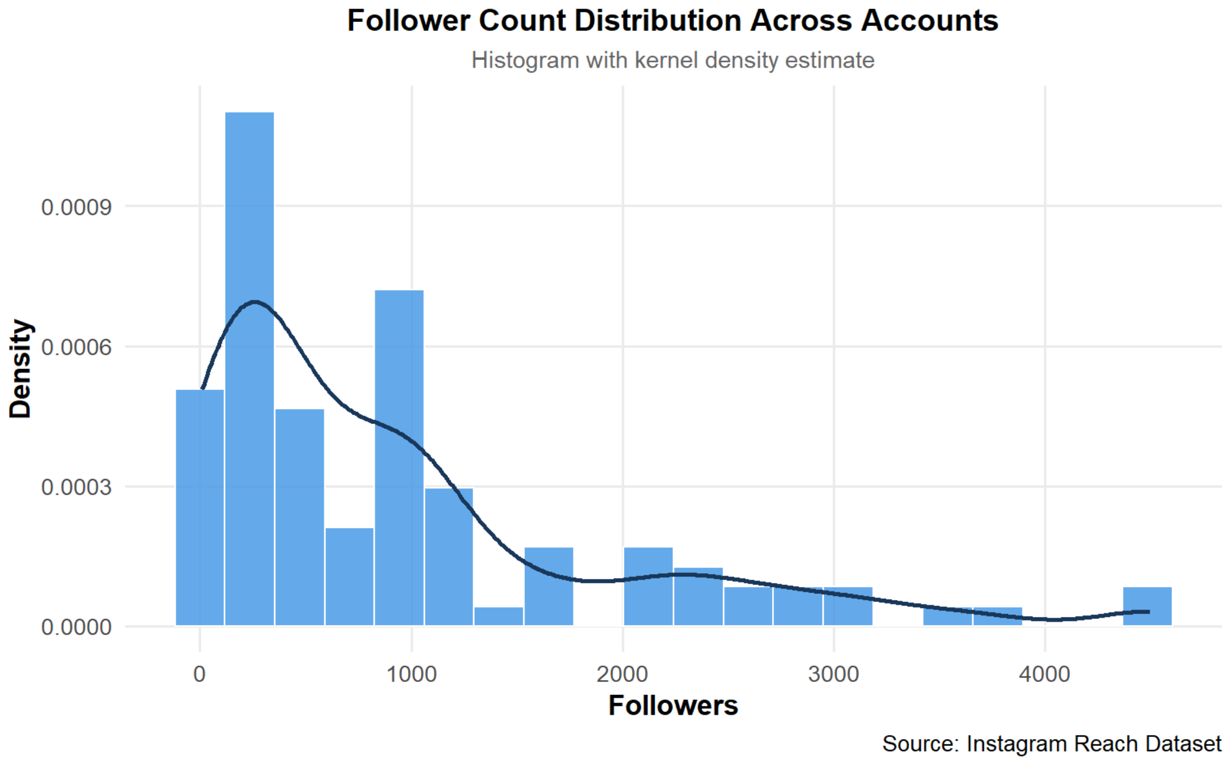
Descriptive Statistics — Numeric Variables

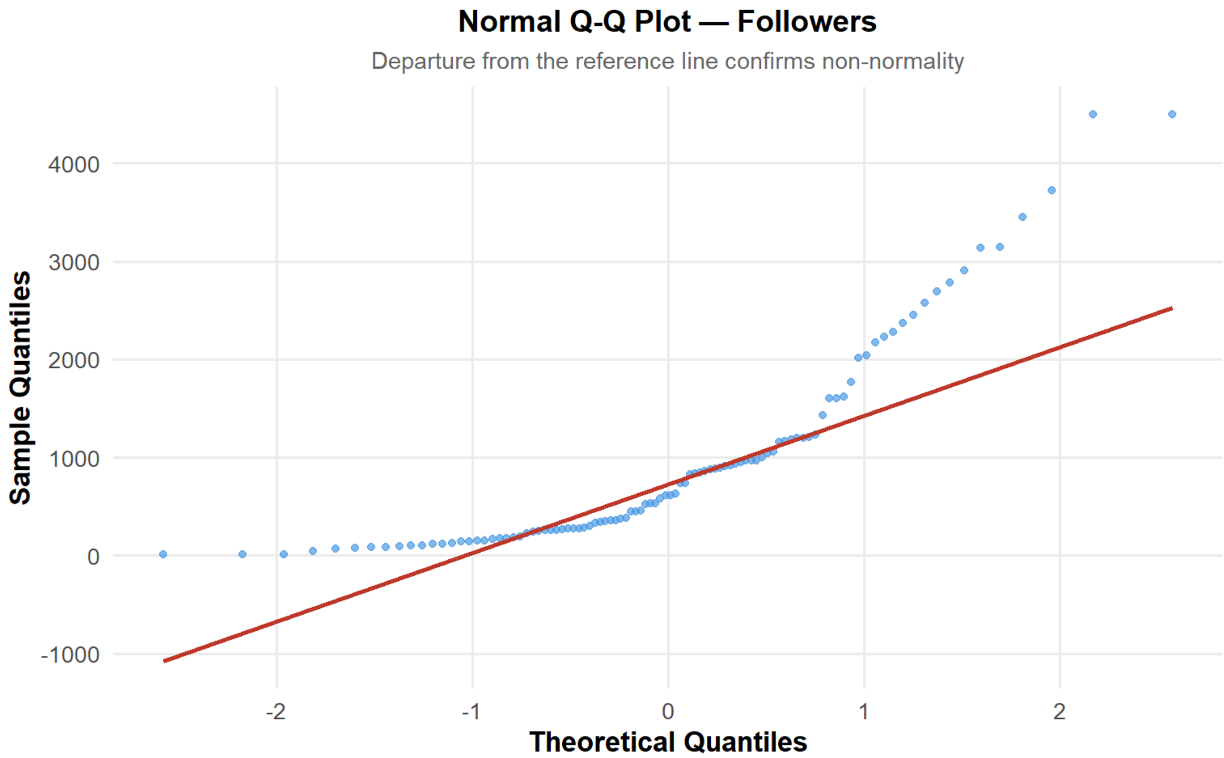
	Variable	Mean	Std. Dev.	Min	Median	Max
Followers	Followers	961.96	1,014.63	11	612	4,496
Hours.since.posted	Hours.since.posted	3.46	3.39	2	2	24
Likes	Likes	46.48	55.09	8	29	349

`Followers` shows the highest variability — its standard deviation is large relative to the mean, driven by a small number of high-follower accounts. `Hours.since.posted` and `Likes` display more moderate spread.

7 Distribution Analysis

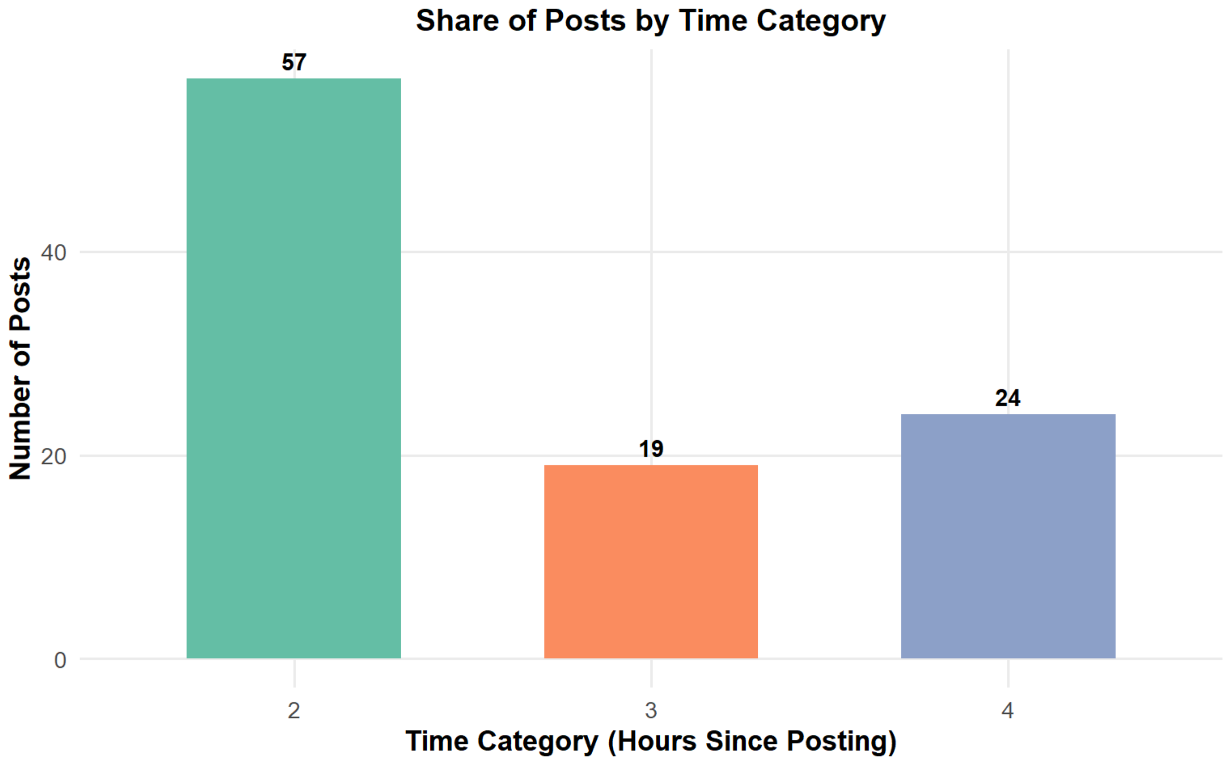
The `Followers` variable is strongly right-skewed, as confirmed by both the histogram and the Q-Q plot. Most accounts have relatively few followers, but several outliers with very high follower counts pull the distribution to the right.

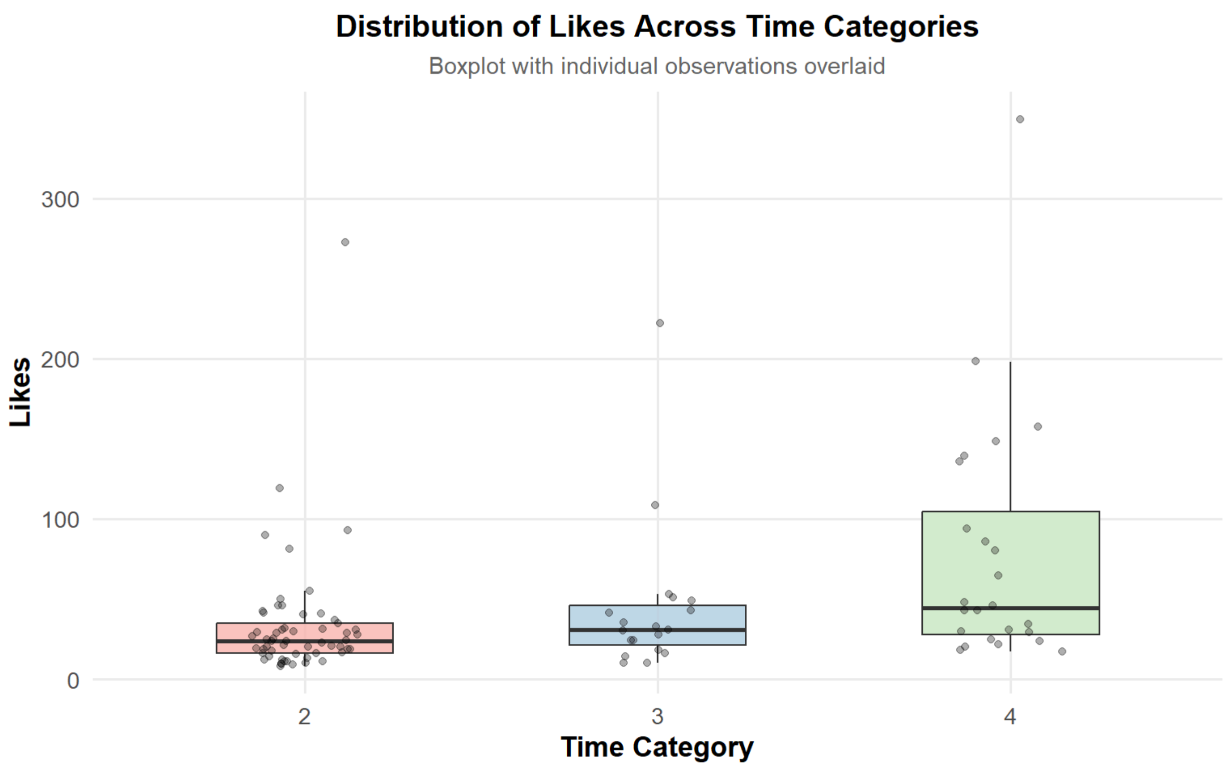




8 Time-Based Engagement Analysis

Posts in the highest time category (4+ hours since posting) tend to accumulate more likes, which is consistent with the general positive relationship between posting age and engagement explored in the next section.



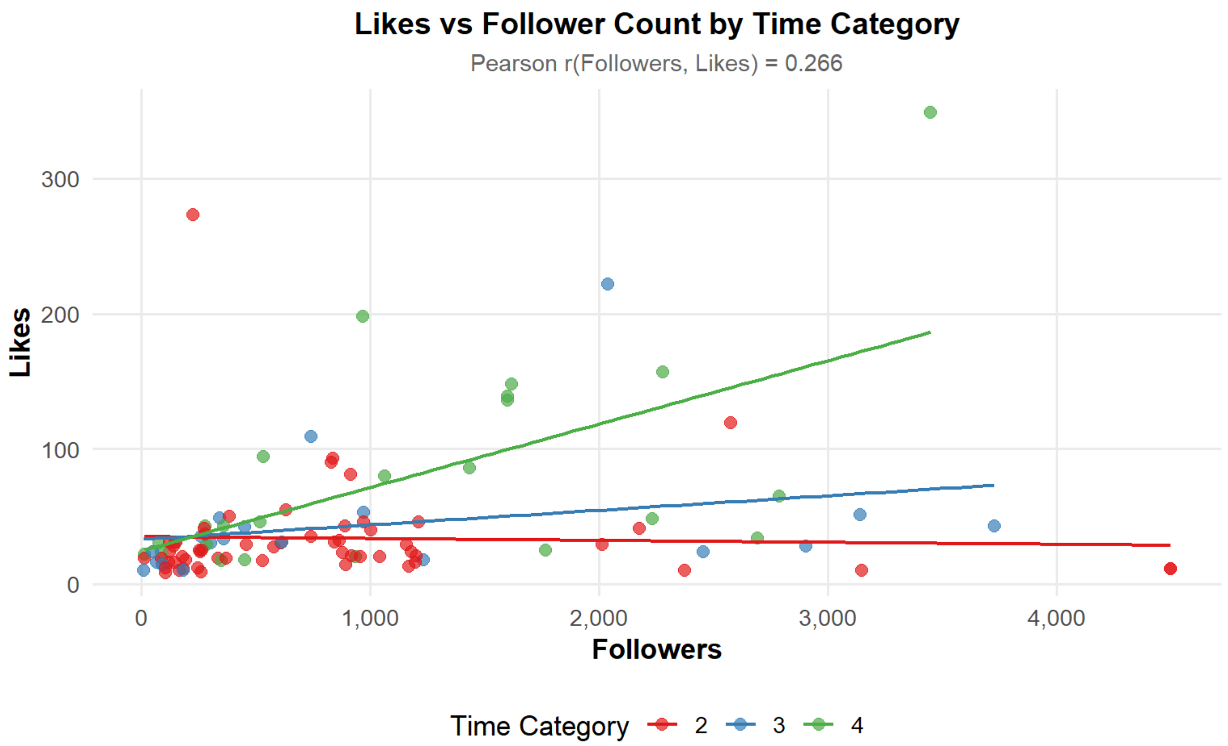


9 Correlation and Bivariate Analysis

Pearson Correlation Matrix — Numeric Variables

	Followers	Hours.since.posted	Likes
Followers	1.000	0.251	0.266
Hours.since.posted	0.251	1.000	0.610
Likes	0.266	0.610	1.000

There is a moderate positive correlation (r ≈ 0.61) between `Hours.since.posted` and `Likes`, suggesting that posts accumulate more likes as time passes. The relationship between `Followers` and `Likes` is weaker (r ≈ 0.27), indicating that follower count is not the primary driver of engagement in this dataset.

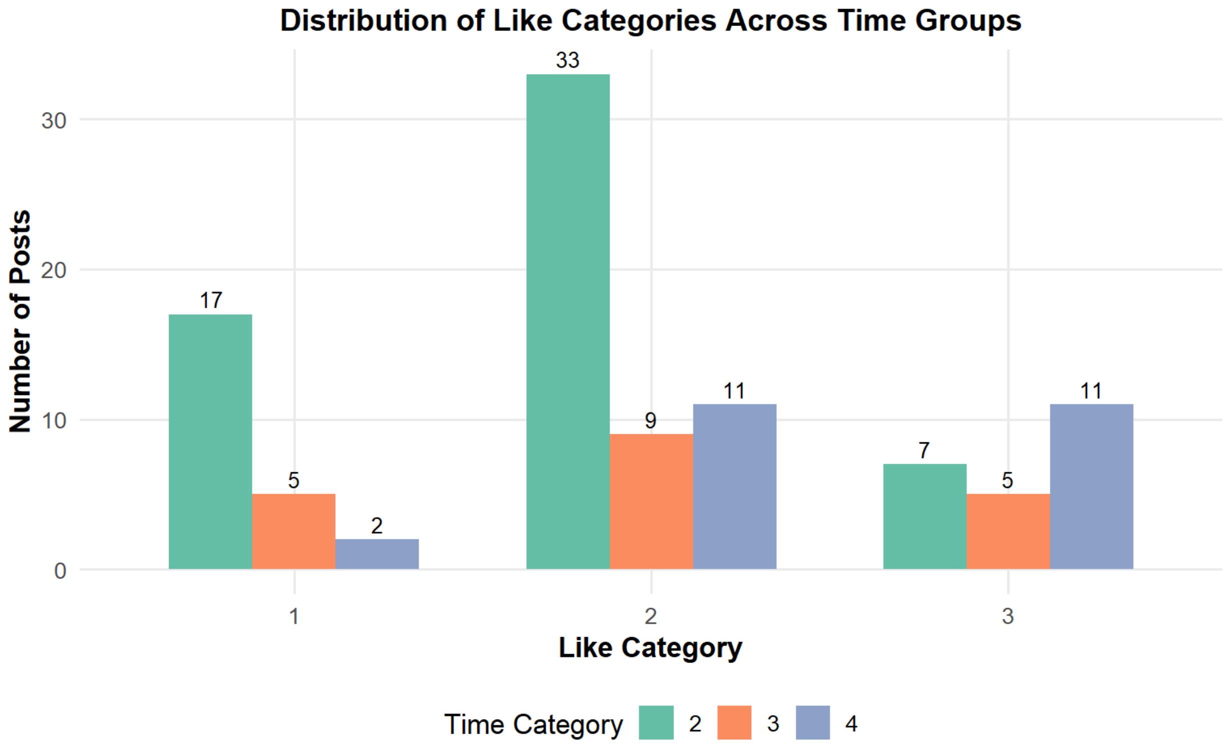


10 Categorical Association Analysis

The contingency table and grouped bar chart below show how like categories distribute across time groups. Posts in the highest time category account for a greater share of high-like outcomes.

Contingency Table: Time Category vs Like Category

Time Category	Low Likes	Medium Likes	High Likes
2	17	33	7
3	5	9	5
4	2	11	11



11 Logistic Regression Modeling

A binary logistic regression model was trained to predict whether a post achieves high engagement (`high_like = 1`), using `Followers` and `Hours.since.posted` as predictors. The dataset was split into a training set (70 posts) and a held-out test set (30 posts) to allow out-of-sample evaluation.

Logistic Regression Coefficients (fitted on full dataset for display)

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	-3.3619	0.6487	-5.1827	0.0000
Followers	0.0004	0.0002	1.7750	0.0759
Hours.since.posted	0.4859	0.1571	3.0925	0.0020

`Hours.since.posted` shows a statistically significant positive effect ($p < 0.01$): as posting age increases, so does the probability of a post being classified as high-engagement. The effect of `Followers` is weaker and less consistent across observations, confirming that posting age is the most statistically significant predictor in this model.

12 Model Evaluation

12.1 Confusion Matrix

Confusion Matrix — Logistic Regression (Test Set, n = 30)

	Low/Medium Engagement	High Engagement

	Low/Medium Engagement	High Engagement
0	22	4
1	1	3

The model identifies 3 true high-engagement posts in the test set. It misses 4 high-engagement cases, which explains the relatively modest recall for that class. This pattern is expected given the class imbalance in the data.

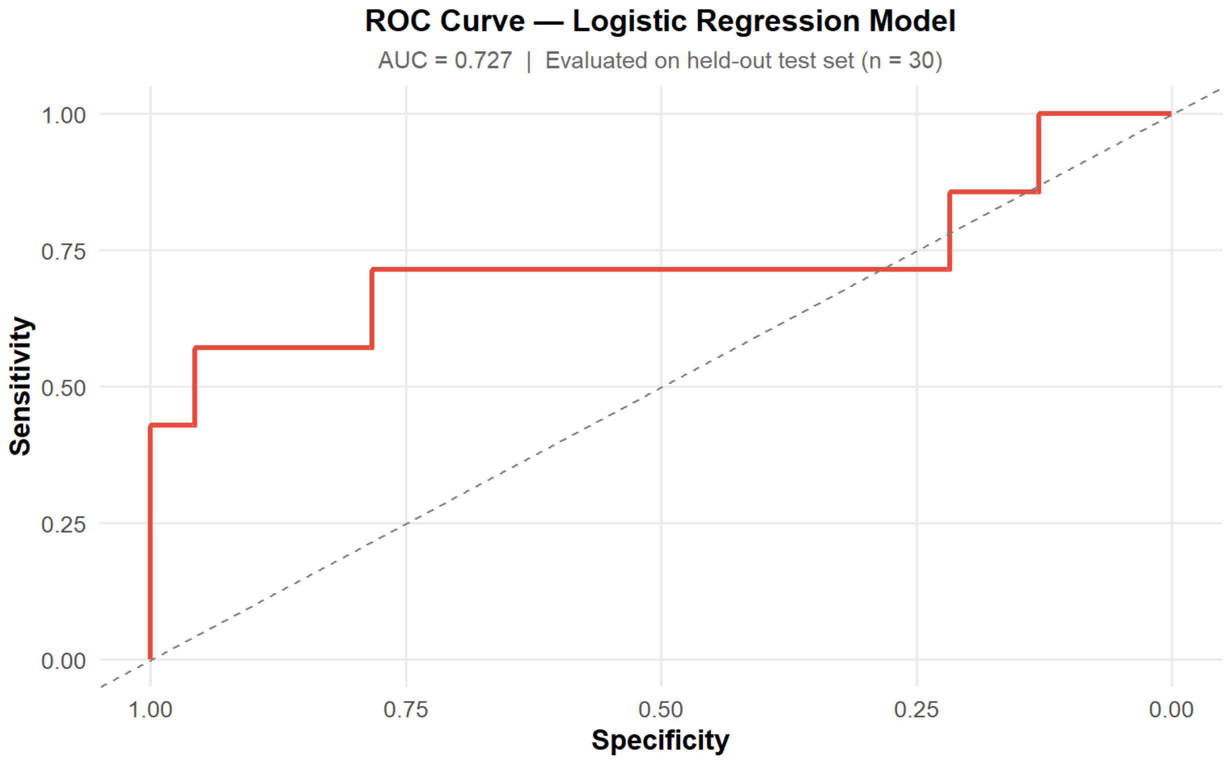
12.2 Performance Metrics

Model Performance Metrics (Test Set)

Metric	Value
Accuracy	0.8333
Precision	0.7500
Recall (Sensitivity)	0.4286
F1-Score	0.5455
AUC	0.7267
Baseline Accuracy	*0.7667*

Although overall accuracy is 83.3%, recall for the high-engagement class remains moderate, which indicates that the model misses a non-trivial share of truly high-engagement posts. The model outperforms a majority-class baseline (76.7% accuracy), confirming that the predictors carry genuine information — but the performance gain is limited by dataset size and class imbalance.

12.3 ROC Curve



An AUC of 0.727 indicates that the model discriminates between high and low/medium engagement posts better than chance, but there remains meaningful room for improvement, particularly in identifying the minority class.

13 Conclusions

- `Hours.since.posted` is the most statistically significant predictor in this model: older posts tend to accumulate more likes.
- The relationship between `Followers` and `Likes` is positive but moderate, suggesting that follower count alone does not reliably predict engagement.
- The `Followers` distribution is strongly right-skewed, driven by a small number of high-follower accounts that inflate the mean.
- Posts in the highest time category show a higher proportion of high-like outcomes, consistent with the regression findings.
- Evaluated on a held-out test set, the logistic regression achieves 83.3% accuracy and AUC 0.727, outperforming a naive baseline but with recall for the high-engagement class remaining the primary performance gap.

14 Limitations

- **Small dataset:** With only 100 observations, model estimates are sensitive to sampling variation and results may not generalize reliably to other contexts.
- **Class imbalance:** Approximately 77% of posts fall in the low/medium engagement category, which inflates accuracy as a standalone metric and limits recall for the minority class.
- **Limited feature set:** Only two predictors are included. Additional features such as caption length, hashtag count, or posting time-of-day could improve predictive performance.
- **Single train/test split:** Performance estimates are based on one 70/30 split. Cross-validation would yield more stable evaluation at the cost of reduced interpretability at this dataset size.